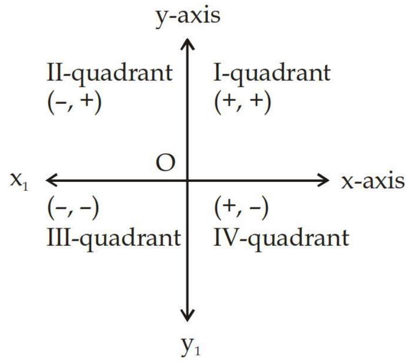
Chapter 18

Co-ordinate Geometry

Cartesian Plane : A plane formed by two perpendicular lines is known as cartesian plane. Here, one line is horizontal and the other line is vertical.

Horizontal line is called x-axis and vertical line is called y-axis.

The point where these lines cross each other is called origin and is denoted by 'O'.



Any point lying on x-axis or y-axis does not lie in any quadrant.

Coordinate of a Point : To locate any point, we use (x, y) here x is known as abscissa and y is known as ordinate.

Abscissa : It is a perpendicular distance from y-axis measured along the x-axis in the positive or negative direction.

Ordinate : It is a perpendicular distance from x-axis measured along the y-axis in the positive or negative direction.

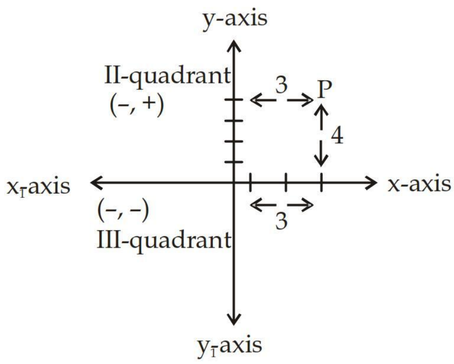
Example: Find the abscissa and ordinate of point P.

Sol. Distance of point P from y-axis = 3

Therefore abscissa is 3.

Distance of point P from x-axis = 4

Therefore ordinate is 4.



(1) Distance Formula :

1. Distance between two points (x_1, y_1), and (x_2, y_2)

$$PQ = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

2. Distance between origin and point P(x, y) is given by = $\sqrt{(x^2 + y^2)}$

Ex. Find the distance between two points A(2, 3) & B(7, 15)

Sol. Distance $= \sqrt{(7-2)^2 + (15-3)^2}$

$= \sqrt{5^2 + 12^2} \quad = \sqrt{25 + 144} \quad = \sqrt{169} \quad = 13$

3. Distance of the straight line $ax + by + c = 0$ from origin $= \left|\frac{c}{\sqrt{a^2+b^2}}\right|$

4. Distance of the straight line $ax + by + c = 0$ from the point $P(x_1, y_1)$ is given by

$$= \left|\frac{ax_1 + by_1 + c}{\sqrt{a^2+b^2}}\right|$$

Ex. Find the distance of straight line $3x + 4y + 15 = 0$ from origin?

Sol. Required distance $= \left|\frac{c}{\sqrt{a^2+b^2}}\right|$

$$= \left|\frac{15}{\sqrt{3^2+4^2}}\right|$$

$$= \frac{15}{5} = 3$$

(2) Section Formula

1. The point $P(x, y)$ dividing the line segment joining the two point $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio $m_1 : m_2$. Then the coordinate of P are

$$x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2} \qquad y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}$$

2. The Point $P(x, y)$ dividing the line segment joining the two points $A(x_1, y_1)$ and $B(x_2, y_2)$ externally in the ratio $m_1 : m_2$. Then the coordinate of point P are

$$x = \frac{m_1x_2 - m_2x_1}{m_1 - m_2} \qquad y = \frac{m_1y_2 - m_2y_1}{m_1 - m_2}$$

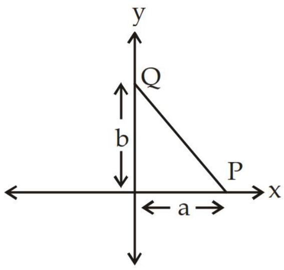
3. If $P(x, y)$ be the midpoint of the line segment joining the two points $A(x_1, y_1)$ and $B(x_2, y_2)$. Then the coordinate of point P are

$$x = \frac{x_1 + x_2}{2} \qquad y = \frac{y_1 + y_2}{2}$$

(3) Intercept Form

If a straight line making intercepts 'a' and 'b' on the x-axis and y-axis respectively. Then the equation of straight line

$$\frac{x}{a} + \frac{y}{b} = 1$$



Slope Intercept Form

1. If a straight line passing through the point $A(x_1, y_1)$ and having a slope 'm'. The equation of straight line is

$$(y - y_1) = m(x - x_1)$$

Two points form

2. If a straight line passing through the two points $A(x_1, y_1)$ and $B(x_2, y_2)$. Then, the equation of a straight line is

$$(y - y_1) = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1) \text{ and the slope is } m = \frac{y_2 - y_1}{x_2 - x_1}$$

Ex. Find the equation of straight line passing through the point (2, 0) and (0, –3)?

Sol. If a line cuts x-axis at (a, 0) and cuts y-axis at (0, b) then the equation is

$$\frac{x}{a}+\frac{y}{b}=1$$

here $a = 2, b = -3$

$$\frac{x}{2}-\frac{y}{3}=1$$

(4) Slope of a line

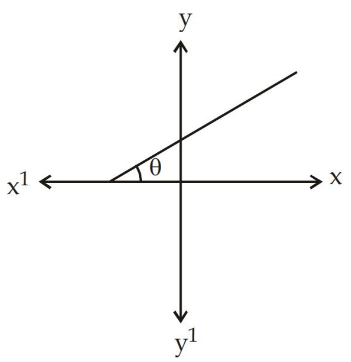
1. If the equation of straight line is $ax + by + c = 0$

then
$$y = -\frac{a}{b}x-\frac{c}{b}$$
$$y = mx + c$$

therefore slope of line $m = \frac{-a}{b}$

and c is constant and intercept on y-axis.

$$m = \tan\theta = \frac{-a}{b}$$



2. Slope of line in terms of co-ordinates. If (x_1, y_1) and (x_2, y_2) are the co-ordinates of any two points on a line, then the slope

$$m = \frac{y_2-y_1}{x_2-x_1} = \frac{\text{Difference of ordinates}}{\text{Difference of abscissa}}$$

If the slope of two lines is m_1 and m_2 then the angle b/w two lines

$$\tan\theta = \frac{m_2-m_1}{1+m_1m_2}$$

The slope of two lines is m_1 and m_2 section for parallel, $m_1 = m_2$

If slope of perpendicular lines m_1 and m_2, then, $m_1m_2 = -1$

The equations of two straight lines are $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$.

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \text{ (Unique solution)}$$

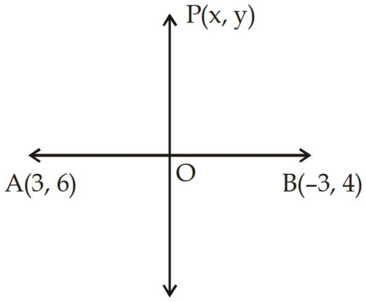
$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \text{ (Infinite many solution)}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \text{ (no solution)}$$

Types of Questions

1. Find the equation of perpendicular bisector of AB where A and B are A(3, 6) and B(–3, 4) respectively.

Sol.



Here $PA = PB$, $PA^2 = PB^2$

$$(x-3)^2 + (y-6)^2 = (x+3)^2 + (y-4)^2$$

$$x^2 - 6x + 9 + y^2 - 12y + 36 = x^2 + 6x + 9 + y^2 - 8y + 16$$

$$12x + 4y - 20 = 0$$

$$3x + y - 5 = 0$$

2. If the point P(4, 3) and Q(x, 5) are on the circle with centre O(2, 3). Find the value of x?

Sol. Here $PO = QO = r$ radius

$$\sqrt{(4-2)^2 + (3-3)^2} = \sqrt{(x-2)^2 + (5-3)^2}$$

$$2 = \sqrt{(x-2)^2 + 4}$$

$$4 = (x-2)^2 + 4$$

$$(x-2)^2 = 0$$

$$x = 2$$

3. What is the distance between parallel lines $2x + 3y + 13 = 0$ and $4x + 6y - 91 = 0$?

Sol. Given lines are

$$2x + 3y + 13 = 0$$

or

$$4x + 6y + 26 = 0$$

$$4x + 6y - 91 = 0$$

Hence, Distance between two parallel lines'

$$ax + by + c = 0$$

and

$$ax + by + d = 0$$

$$\text{Distance} = \left|\frac{d-c}{\sqrt{a^2+b^2}}\right|$$

$$= \frac{117}{\sqrt{52}} = \frac{13 \times 9}{2\sqrt{13}}$$

$$= \frac{9\sqrt{13}}{2} \text{ units}$$

4. Find the coordinates of the point of intersection of the medians of a triangle with vertices A(0, 6), B(5, 3) and C (7, 3).

Sol. We known that the point of intersection of medians is centroid.

Coordinate of centroid

$$= \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

$$= \left(\frac{12}{3}, \frac{12}{3}\right)$$

$$= (4, 4)$$

5. Find the angle between the lines represented by the equation $2y - \sqrt{12}x - 9 = 0$ and $\sqrt{3}y - x + 7 = 0$.

Sol.

$$2y - \sqrt{12}x - 9 = 0$$

$$2y = \sqrt{12}x + 9$$

$$y = \frac{\sqrt{12}}{2}x + \frac{9}{2}$$

Here, $m_1 = \frac{\sqrt{12}}{2} = \sqrt{3}$

and $\sqrt{3}y - x + 7 = 0$

$$\sqrt{3}y = x - 7$$

$$y = \left(\frac{1}{\sqrt{3}}\right)x - \frac{7}{\sqrt{3}}$$

$$m_2 = \frac{1}{\sqrt{3}}$$

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$= \frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{1 + \sqrt{3}\cdot\frac{1}{\sqrt{3}}} = \frac{\frac{2}{\sqrt{3}}}{2}$$

$$\tan\theta = \frac{1}{\sqrt{3}}$$

So, $\theta = 30°$

6. Find the equation of a line passing through the point (5, 3) and parallel to the line $2x - 5y + 3 = 0$.

Sol. $2x - 5y + 3 = 0$

$y = \left(\frac{2}{5}\right)x + \frac{3}{5}$ Its slope $m_1 = \frac{2}{5}$

Slope of line parallel to the given lines is $\frac{2}{5}$

$$y = \frac{2}{5}x + c$$

The line is passing through (5, 3)

$$3 = \frac{2}{5} \times 5 + c$$

$$c = 3 - 2 = 1$$

The equation of required line is

$$y = \frac{2}{5}x + 1$$

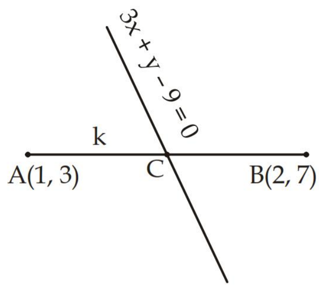
$$2x - 5y + 5 = 0$$

7. Find the ratio in which the line $3x + y - 9 = 0$ divides the segment joining the point (1, 3) and (2, 7).

Sol. Required ratio= K : 1

Co-ordinate of point c.

$$= \left(\frac{2k+1}{k+1}, \frac{7k+3}{k+1}\right)$$



and c also lies on $3x + y - 9 = 0$

$$3\left(\frac{2k+1}{k+1}\right) + \frac{7k+3}{k+1} - 9 = 0$$

$$6k + 3 + 7k + 3 - 9k - 9 = 0$$

$$4k = 3$$

$$\frac{k}{1} = \frac{3}{4}$$

$$K : 1 = 3 : 4$$

Foundation

Questions

1. If point (a, a + 2) lies on the line $y = 3x + 5$, then distance of the point from y-axis is:

   (a) $\frac{3}{2}$ (b) 3

   (c) $\frac{1}{2}$ (d) $\frac{7}{2}$

2. What is the difference between distances of mid point of points (– 3, 5) and (7, – 6) from x-axis and y-axis?

   (a) $\frac{5}{2}$ (b) $\frac{3}{2}$

   (c) $\frac{5}{4}$ (d) $\frac{3}{4}$

3. The length intercepted by the straight line $12x - 9y = 108$ between the coordinate axes is:

   (a) 12 unit (b) 18 unit

   (c) 15 unit (d) 9 unit

4. The length intercepted by the straight line $y = mx + c$ between the coordinate axes is:

   (a) $\frac{c}{m}\sqrt{1+m^2}$ (b) $\frac{c}{m}$

   (c) $\sqrt{c^2 + m^2}$ (d) None of these

5. Area of triangle formed by the straight line $8x - 3y + 24 = 0$ and coordinate axes is:

   (a) 24 sq. unit (b) 12 sq. unit

   (c) 6 sq. unit (d) 18 sq. unit

6. Area of triangle formed by straight lines $4x - 3y + 4 = 0$, $4x + 3y - 20 = 0$ and x-axis is:

   (a) 3 sq. unit (b) 6 sq. unit

   (c) 12 sq. unit (d) 24 sq. unit

7. Area of triangle formed by straight lines $3x - 4y = 0$, $x = 4$ and x-axis is:

   (a) 6 sq. unit (b) 8 sq. unit

   (c) 12 sq. unit (d) 16 sq. unit

8. Area of triangle formed by straight lines $3x + 4y = 24$, $x = 8$ and $y = 6$ is:

   (a) 12 sq. unit (b) 24 sq. unit

   (c) 48 sq. unit (d) None of these

9. Which of the following system of equations has unique solutions?

   (a) $3x + 4y = 11, 6x + 8y = 15$

   (b) $x + 2y = 3, 2x + 4y = 7$

   (c) $y = 3x - 4, 3x - y - 4 = 0$

   (d) $4x + 3y = 5, 4x - 3y = 5$

10. For what values of k straight lines $2x - ky + 3 = 0$ and $3x + 2y - 1 = 0$ are parallel?
(a) $\frac{4}{3}$ (b) $-\frac{4}{3}$
(c) $\frac{2}{3}$ (d) $-\frac{2}{3}$

11. The value of K for which system of equation $5x + 2y = K$ and $10x + 4y - 3 = 0$ has infinitely many solution is:
(a) $\frac{3}{2}$ (b) $\frac{1}{3}$
(c) 6 (d) $\frac{1}{6}$

12. Ratio of area of triangle formed by straight lines $2x + 3y = 4$ and $3x - y + 5 = 0$ with x-axis and y-axis is:
(a) 1 : 2 (b) 2 : 1
(c) 4 : 1 (d) None of these

13. The point (7, – 5) lies in the quadrant:
(a) First (b) Second
(c) Third (d) Fourth

14. The distance between the points A (b, o) and B (0, a) is:
(a) $\sqrt{a^2 - b^2}$ (b) $\sqrt{a^2 + b^2}$
(c) $\sqrt{a + b}$ (d) a + b

15. The co-ordinates of a point situated on y-axis at a distance of 7 units from x-axis is:
(a) (0, 7) (b) (7, 0)
(c) (7, 7) (d) (– 7, 7)

16. The slope of the line $3x + 7y + 8 = 0$ is:
(a) 3 (b) 7
(c) $-\frac{3}{7}$ (d) $\frac{3}{7}$

17. The equation of a line parallel to y-axis at a distance of 5 units to the left of y-axis, is:
(a) $y = -5$ (b) $x = -5$
(c) $x + 5y = 0$ (d) $y + 5x = 0$

18. The equation of a line passing through the points A(0, – 3) and B (– 5, 2) is:
(a) $x + y + 3 = 0$ (b) $x + y - 3 = 0$
(c) $x - y + 3 = 0$ (d) $x - y - 3 = 0$

19. If the points P(2, 3), Q(5, a) and R (6, 7) are collinear, the value of a is:
(a) $\frac{5}{2}$ (b) $-\frac{4}{3}$
(c) 6 (d) 5

20. The ratio in which the line segment joining P(– 3, 7) and Q(7, 5) is divided by y-axis is:
(a) 3 : 7 (b) 4 : 7
(c) 3 : 5 (d) 4 : 5

21. The value of a so that the lines $x + 3y - 8 = 0$ and $ax + 12y + 5 = 0$ are parallel is:
(a) 0 (b) 1
(c) 4 (d) – 4

22. The co-ordinates of the point P which divides the line joining A(3, – 2) and $B\left(\frac{11}{2}, \frac{21}{2}\right)$ in the ratio 2 : 3 are:
(a) (4, 3) (b) (4, 5)
(c) $\left(4, \frac{5}{2}\right)$ (d) $\left(\frac{3}{2}, \frac{7}{2}\right)$

23. The angle between two legs of a compass is 60° and length of each leg is 10 cm. The distance between end points of the leg is:
(a) 5 cm (b) 10 cm
(c) $5\sqrt{3}$ cm (d) $10\sqrt{3}$ cm

24. O is a point on the line LM. A line ON is drawn which is neither coincident with OL nor with OM. If $\angle$MON is one third of $\angle$LON, then $\angle$MON is equal to:
(a) 45° (b) 60°
(c) 75° (d) 80°

25. Let the vertices of a triangle ABC be (4, 3), (7, – 1), (9, 3) then the triangle is:
(a) Scalene (b) Isosceles
(c) Equilateral (d) None of these

26. Find the co-ordinates of the point which divides the line joining the points (2, 4) and (6, 8) externally in the ratio 5 : 3?
(a) (5, 6) (b) (12, 14)
(c) (3, 8) (d) (2, 7)

27. The coordinates of the vertices of a triangle are A(3, 1), B(2, 3) and C(– 2, 2). Find the coordinates of the centroid of the triangle ABC:
(a) (1, 2) (b) (2, 3)
(c) (4, 5) (d) (5, 6)

28. A(– 2, – 1), B(1, 0), C(4, 3) and D(1, 2) are the four points of a quadrilateral ABCD. The quadrilateral is a:
(a) Square (b) Rhombus
(c) Parallelogram (d) None of (a), (b), (c)

29. Find the slope of the line joining the points (7, 5) and (9, 7):
(a) 1 (b) 2
(c) $\frac{1}{2}$ (d) 3

30. Three points A(1, – 2), B(3, 4) and C(4, 7) form:
(a) a straight line
(b) an equilateral triangle
(c) a right angled triangle
(d) None of (a), (b), (c)

Moderate

1. The slope of the line joining P(– 4, 7) and Q(2, 3) is:
(a) $-\frac{2}{3}$ (b) $\frac{2}{3}$
(c) $-\frac{3}{2}$ (d) $\frac{3}{2}$

2. The area of the triangle whose vertices are P(4, 5), Q(– 3, 8) and R (3, – 4), (in square units) is:
(a) 66 (b) $16\frac{1}{2}$
(c) 33 (d) 35

3. The co-ordinates of the centroid of ΔPQR with vertices P(– 2, 0), Q(9, – 3) and R (8, 3) is:
(a) (1, 0) (b) $\left(\frac{19}{3}, 0\right)$
(c) (0, 5) (d) (5, 0)

4. The angle which the line joining the points $\left(\sqrt{3}, 1\right)$ and $\left(\sqrt{15}, \sqrt{5}\right)$ makes with x-axis is:
(a) 30° (b) 45°
(c) 60° (d) 90°

5. If the sum of the square of the distance of the point (x, y) from the point (a, 0) and (– a, 0) is $2b^2$, then:
(a) $x^2 + a^2 = b^2 + y^2$ (b) $x^2 + a^2 = 2b^2 - y^2$
(c) $x^2 - a^2 = b^2 + y^2$ (d) $x^2 + a^2 = b^2 - y^2$

6. Two vertices of a triangle PQR are P(– 1, 0) and Q(5, –2) and its centroid is (4, 0). The co-ordiantes of R is:
(a) (8, – 2) (b) (8, 2)
(c) (– 8, 2) (d) (– 8, – 2)

7. The ratio in which the line segment joining A(3, – 5) and B(5, 4) is divided by x-axis is:
(a) 4 : 5 (b) 5 : 4
(c) 5 : 7 (d) 6 : 5

8. The value of a so that line joining P(– 2, 5) and Q(0, – 7) and the line joining A(– 4, – 2) and B(8, a) are perpendicular to each other is:
(a) – 1 (b) 5
(c) 1 (d) 0

9. The angle between the lines represented by the equations $2y - \sqrt{12}x - 9 = 0$ and $\sqrt{3}y - x + 7 = 0$, is:
(a) 30° (b) 45°
(c) 60° (d) $22\frac{1}{2}°$

10. Area of triangle formed by straight lines 2x + 3y = 5 and y = 3x – 13 with x-axis is:
(a) $\frac{11}{3}$ sq. unit (b) $\frac{22}{3}$ sq. unit
(c) $\frac{11}{6}$ sq. unit (d) $\frac{11}{12}$ sq. unit

11. Area of triangle formed by straight lines 4x – y = 4, 3x + 2y = 14 and y-axis is:
(a) $\frac{11}{2}$ sq. unit (b) $\frac{11}{4}$ sq. unit
(c) 22 sq. unit (d) 11 sq. unit

12. Area of triangle formed by straight lines 2x – 3y + 6 = 0, 2x + 3y – 18 = 0 and y – 1 = 0 is:
(a) 27 sq. unit (b) $\frac{27}{2}$ sq. unit
(c) 9 sq. unit (d) None of these

13. Area of quadrilateral formed by straight lines x + y = 2, 3x + 4y = 24 and coordinate axes is:
(a) 22 sq. unit (b) 26 sq. unit
(c) 44 sq. unit (d) 11 sq. unit

14. Area of quadrilateral formed by straight lines 2x = – 5, 2y = 3, x + 1 = 0 and y + 2 = 0 is:
(a) $\frac{21}{2}$ sq. unit (b) $\frac{21}{4}$ sq. unit
(c) $\frac{21}{8}$ sq. unit (d) $\frac{21}{16}$ sq. unit

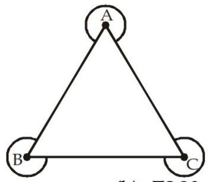
15. Area of triangle formed by straight lines x – y = 0, x + y = 0 and 2x = 5 is:
(a) 25 (b) $\frac{25}{2}$
(c) $\frac{25}{4}$ (d) None of these

16. Values of a and b so that system of equations 2x + 3y = 7 and 2ax + (a + b) y = 28 has infinitely many solutions are:
(a) a = 4, b = 8 (b) a = 8, b = 4
(c) a = – 4, b = – 8 (d) a = – 8, b = – 4

17. Area enclosed by equation y = |x| – 1 and y = 1 – |x| is:
(a) 2 (b) 4
(c) 8 (d) 16

18. Length of line segment AB is 2 unit. It is divided by point C such that $AC^2 = AB \times CB$, the length of CB is:
(a) $3+\sqrt{5}$ unit (b) $3-\sqrt{5}$ unit
(c) $2-\sqrt{5}$ unit (d) $\sqrt{3}$ unit

19. In the figure given below sum of angles around the point A, B, C excluded internal angles of the triangle is:



(a) 360° (b) 720°
(c) 900° (d) 1000°

20. If the inclination of a line joining the points A (x, – 3) and B (2, 5) is 135°, then x = ?
(a) 10 (b) 15
(c) 20 (d) 25

21. If the graph of the equation 2x + 3y = 6 form a triangle with coordinates axis, then the area of triangle will be:
(a) 2 sq. units (b) 3 sq. units
(c) 6 sq. units (d) 1 sq. units

22. The value of k for which the lines x + 2y = 9 and kx + 4y = – 5 are parallel, is:
(a) k = 2 (b) k = 1
(c) k = – 1 (d) k = 3

23. Find the equation of perpendicular bisector of the line made by joining the points (1, 1) and (3, 5)?
(a) x + 2y + 8 = 0 (b) x – 2y + 8 = 0
(c) x – 2y – 8 = 0 (d) x + 2y – 8 = 0

24. If three vertices of a parallelogram are A(3, 5) B(– 5, – 4) and C(7, 10), then fourth vertex is:
(a) (10, 19) (b) (15, 19)
(c) (19, 10) (d) (10, 15)

25. Are the points (1, 7), (4, 2), (– 1, – 1) and (– 4, 4) vertices of a square? If yes, what is the length of the side of square?
(a) 34 (b) $\sqrt{34}$
(c) $\sqrt{68}$ (d) 68

26. If P(a, 0), Q(0, b) and R(1, 1) are collinear. Then, find the value of $\frac{1}{a}+\frac{1}{b}$.
(a) 2 (b) 1
(c) – 1 (d) 0

Difficult

1. If A(– 5, 7) B(– 4, – 5) C(– 1, – 6) and D(4, 5) are the vertices of a quadrilateral, then find the area of the quadrilateral ABCD?
(a) 72 sq. units (b) 80 sq. units
(c) 90 sq. units (d) 92 sq. units

2. The sides PQ, QR, RS and SP of a quadrilateral have the equations x + 2y = 3, x = 1, x – 3y = 4, 5x + y + 12 = 0 respectively, then the angle between the diagonals PR and QS is:
(a) 30° (b) 45°
(c) 60° (d) 90°

3. If P_1 and P_2 be perpendicular from the origin upon the straight lines x secθ + y cosecθ = a and x cosθ – y sin θ = a cos 2θ respectively, then the value of $4P_1^2 + P_2^2$ is:
(a) a^2 (b) $2a^2$
(c) $\sqrt{2}\,a^2$ (d) $3a^2$

4. Find the area of the quadrilateral whose coordinates of angular points taken in order are (1, 1), (3, 4), (5, – 2) and (4, – 7)?
(a) 20.5 (b) 41
(c) 82 (d) 61.5

5. The straight line joining (1, 2) and (2, – 2) is perpendicular to the line joining (8, 2) and (4, p). What will be the value of p?
(a) – 1 (b) 1
(c) 3 (d) None of these

6. Find the equation of the straight line passing through the origin and the point of intersection of the lines $\frac{x}{a}+\frac{y}{b}=1$ and $\frac{x}{b}+\frac{y}{a}=1$?
(a) y = x (b) y = – x
(c) y = 2x (d) y = – 2x

7. Find the equation of the line joining the points of intersection of 2x + y = 4 with x – y + 1 = 0 and 2x – y – 1 = 0 with x + y – 8 = 0?
(a) 2x + 3y + 6 = 0 (b) 3x + 2y + 12 = 0
(c) 3x – 2y + 1 = 0 (d) None of (a), (b), (c)

8. Find the equation of the line which pass through the point (4, 5) which make an acute angle 45° with the line 2x – y + 7 = 0?
(a) x – 2y = 0 (b) 7x + 5y – 3 = 0
(c) 3x + y + 8 = 0 (d) x – 3y + 11 = 0

9. Find the equation of the straight line which passes through the point of intersection of the straight lines 3x – 4y + 1 = 0 and 5x + y – 1 = 0 and cuts off equal intercepts from the axis?

(a) $32x + 32y + 11 = 0$ (b) $23x + 23y = 11$
(c) $9x + 18y + 5 = 0$ (d) None of these

10. Find the coordinates of the orthocentre of the triangle whose vertices are (1, 2), (2, 3) and (4, 3)?
(a) (2, 5) (b) (3, 4)
(c) (1, 6) (d) None of these

11. The area of a triangle is 5. Two of its vertices are (2, 1) and (3, – 2). The third vertex lies on $y = x + 3$. Find the third vertex?
(a) $\left(\frac{2}{7}, \frac{13}{2}\right)$ (b) $\left(\frac{7}{2}, \frac{13}{2}\right)$
(c) $\left(\frac{9}{2}, \frac{13}{2}\right)$ (d) $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$

12. A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L and coordinate axes is 5. Find the equation of the line?
(a) $x + 5y = \pm 5\sqrt{2}$ (b) $x - 3y = 0$
(c) $2x + y = 0$ (d) $x + 4y = 5\sqrt{2}$

13. Find the equation of the straight line which passes through the point of intersection of the straight lines $x + y = 8$ and $3x - 2y + 1 = 0$ and is parallel to the straight line joining the points (3, 4) and (5, 6)?
(a) $x - y + 2 = 0$ (b) $x + y - 2 = 0$
(c) $3x - 4y + 8 = 0$ (d) None of these

14. Find the equation of the line which passes through the point of intersection of the lines $x + 2y - 3 = 0$ and $4x - y + 7 = 0$ and is parallel to the line $y - x + 10 = 0$?
(a) $2x + 2y + 5 = 0$ (b) $3y - 3x = 10$
(c) $3x + 2y - 8 = 0$ (d) None of these

15. A straight line $\frac{x}{a} - \frac{y}{b} = 1$ passes through the point (8, 6) and cuts off a triangle of area 12 units from the axes of coordinates. Find the equation of the straight line?
(a) $3x - 2y = 12$ (b) $4x - 3y = 12$
(c) $3x - 8y + 24 = 0$ (d) both (a) and (c)

16. Two vertices of a triangle ABC are B(5, – 1) and C(– 2, 3). If the orthocentre of the triangle is the origin, find the third vertex?
(a) $\left(\frac{7}{2}, \frac{13}{2}\right)$ (b) $\left(\frac{3}{2}, \frac{11}{2}\right)$
(c) (– 4, – 7) (d) None of these

17. Find the equations of the bisectors of the angle between the straight line $3x + 4y + 2 = 0$ and $5x - 12y - 6 = 0$?
(a) $8x + y + 7 = 0$ (b) $16x - 2y - 1 = 0$
(c) $x + 8y + 4 = 0$ (d) both (b) and (c)

18. A straight line passes through the points (a, 0) and (0, b). The length of the line segment contained between the axis is 13 and the product of the intercepts on the axes is 60. Find the equation of the straight line?
(a) $5x + 12y = 60$ (b) $7x - 12y = 50$
(c) $5x + 12y + 60 = 0$ (d) both (a) and (c)

19. Find the equation of the straight line which passes through the point (3, 4) and has intercepts on the axes such that their sum is 14?
(a) $4x + 3y = 24$ (b) $x + y = 7$
(c) $3x + 7y = 43$ (d) both (a) and (b)

20. Point A(4, – 1), B(6, 0), C(7, 2) and D(5, 1) are the vertices of the quadrilateral which is a?
(a) Square (b) Rectangle
(c) Rhombus (d) None of (a), (b), (c)

Previous Year (Memory Based)

1. The graph of the equation $2x - 3y = 6$ intersects the y-axis at the point?
(a) (– 2, 0) (b) (0, – 2)
(c) (2, 3) (d) (2, – 3)

2. The equation of line whose graph passes through the origin out of the given equations $2x + 3y = 2$, $2x - 3y = 3$, $-2x + 3y = 5$ and $2x + 3y = 0$, is:
(a) $2x - 3y = 3$ (b) $-2x + 3y = 5$
(c) $2x + 3y = 0$ (d) $2x + 3y = 2$

3. The graph of the linear equations $3x + 4y = 24$ is a straight line intersecting x-axis and y-axis at the points A and B, respectively. P(2, 0) and $Q\left(0, \frac{3}{2}\right)$ are two points on the sides OA and OB respectively of ΔOAB, where O is the origin of the coordinate system. Given that AB = 10 cm, then PQ is equal to:
(a) 20 cm (b) 2.5 cm
(c) 40 cm (d) 5 cm

4. The length of the intercept of the graph of the equation $9x - 12y = 108$ between the two axes is:
(a) 15 units (b) 9 units
(c) 12 units (d) 18 units

5. Area of the triangle formed by the graph of the straight lines $x - y = 0$, $x + y = 2$ and the x-axis is:
(a) 1 sq. unit (b) 2 sq. units
(c) 4 sq. units (d) None of these

6. The area of the triangle formed by the graphs of $3x + 4y = 12$, x-axis and y-axis (in sq. units) is:
(a) 6 (b) 8
(c) 4 (d) 12

7. The x-intercept of the graph of $7x - 3y = 2$ is:
(a) $\frac{3}{4}$ (b) $\frac{3}{7}$
(c) $\frac{2}{5}$ (d) $\frac{2}{7}$

8. If $2x + y = 6$ and $x = 2$ are two linear equations, then graph of two equations meet at a point?
(a) (2, 0) (b) (0, 2)
(c) (2, 2) (d) (1, 2)

9. The x-intercept of the graph of $5x + 6y = 30$ is:
(a) 4 units (b) 5 units
(c) 6 units (d) 15 units

10. The lines $2x + y = 5$ and $x + 2y = 4$ intersect at the point?
(a) (1, 2) (b) (2, 1)
(c) $\left(\frac{5}{2}, 0\right)$ (d) (0, 2)

11. The area bounded by the lines $x = 0$, $y = 0$, $x + y = 1$ and $2x + 3y = 6$ (in sq. units) is:
(a) 2 (b) $2\frac{1}{3}$
(c) $2\frac{1}{2}$ (d) 3

12. The area of the triangle formed by the lines $5x + 7y = 35$ and $4x + 3y = 12$ and x-axis is:
(a) $\frac{160}{13}$ sq. units (b) $\frac{150}{13}$ sq. units
(c) $\frac{140}{13}$ sq. units (d) 10 sq. units

13. For what value of k system of equation $x + 2y = 5$, $3x + ky + 15 = 0$ doesnot have any solution?
(a) 2 (b) – 2
(c) 6 (d) – 6

14. In the x-y coordinate system, if (a, b) and (a + 3, b + k) are two points on the line defined by the equation $x = 3y - 7$, then k =
(a) 9 (b) 3
(c) $\frac{7}{3}$ (d) 1

15. The length of the portion of the straight line $8x + 15y = 120$ intercepted between the axes is:
(a) 14 units (b) 15 units
(c) 16 units (d) 17 units

16. A triangle is formed by the x-axis and the lines $2x + y = 4$ and $x - y + 1 = 0$ as three sides. Taking the side along x-axis as its base, the corresponding altitude fo the triangle is:
(a) 2 unit (b) 3 unit
(c) $\sqrt{5}$ unit (d) 1 unit

17. The x-intercept of the graph of $5x - 4y = 20$ is:
(a) 4 units (b) 5 units
(c) 9 units (d) 1 unit

18. Area of the trapezium formed by x-axis, y-axis and the lines $3x + 4y = 12$ and $6x + 8y = 60$ is:
(a) 31.5 sq. units (b) 48 sq. units
(c) 36.5 sq. units (d) 37.5 sq. units

19. Area of the triangle formed by the graph of the line $2x - 3y + 6 = 0$ along with the coordinate axes is:
(a) $\frac{3}{2}$ sq. units (b) 3 sq. units
(c) 6 sq. units (d) $\frac{1}{2}$ sq. units

20. If (4, 5) is one end of the diameter of a circle with radius 5 units and centre on x-axis, then the centre of the circle is ______.
(a) (0, 4) (b) (– 4, 0)
(c) (4, 0) (d) (0, – 4)

21. If (0, 0), (3, 3) and (k, 0) form a right angled isosceles triangle, then k could be ______.
(a) 4 (b) 2
(c) – 3 (d) 6

22. If the points (a, 4), (2, 2) and (5, 5) are collinear, then a =
(a) 1 (b) 2
(c) 3 (d) 4

23. If (3, 3), (– 1, – 1) are two vertices of an equilateral triangle, then the third vertex can be:
(a) (2, 2) (b) (– 5, – 5)
(c) (– 2, – 2) (d) $\left(-2\sqrt{3}+1, 2\sqrt{3}+1\right)$

24. For what value of k are the graphs of $(k - 2)x + 3y - 4 = 0$ and $(k - 3)x - 2y + 5 = 0$ parallel?
(a) $-\frac{13}{5}$ (b) $\frac{13}{5}$
(c) $\frac{11}{5}$ (d) $-\frac{11}{5}$

25. The value of k for which the lines $2x + 3y - 6 = 0$ and $x + ky + 1 = 0$ are perpendicular to each other is ______.
(a) $-\frac{2}{3}$ (b) 3
(c) 4 (d) $-\frac{4}{3}$

26. If the lines x + y + 3 = 0, x – y – 1 = 0 and ax + y + 4 = 0 are concurrent, then a =
(a) – 1 (b) 3
(c) 2 (d) 1

27. The distance of the line 4x + 3y – 9 = 0 from the origin is:
(a) $\frac{9}{25}$ (b) $\frac{9}{5}$
(c) $\frac{81}{5}$ (d) $\frac{81}{25}$

28. The equation of the graph passing through the point (1, 1) which makes an angle of 60° with the line $3x - y\sqrt{3} + 7 = 0$ can be _____.
(a) y – 2 = 0 (b) y – 3 = 0
(c) y – 1 = 0 (d) y + 1 = 0

29. Distance of the point (2, 3) from the line 2x – 3y + 9 = 0 measured along the line x – y + 1 = 0 is _____.
(a) $4\sqrt{2}$ (b) $\sqrt{8}$
(c) $\sqrt{2}$ (d) $5\sqrt{2}$

Foundation

Solutions

1. **(a);** a + 2 = 3a + 5

$\Rightarrow \; a = \frac{-3}{2}$

So, the given point is $\left(\frac{-3}{2}, \frac{1}{2}\right)$

$\Rightarrow$ The distance of the point from y-axis $= \frac{3}{2}$.

2. **(b);** mid-point of given points is:

$\left(\frac{-3+7}{2}, \frac{5-6}{2}\right) \equiv \left(2, \frac{-1}{2}\right)$

The distances of the above point from x and y axes are $\frac{1}{2}$ and 2 respectively.

difference $= 2 - \frac{1}{2} = \frac{3}{2}$

3. **(c);** 12x – 9y = 108
At x = 0, y = – 12
At y = 0, x = 9
So, the given line meets the coordinate axes at (9, 0) and (0, – 12).

So, the required length $= \sqrt{(9-0)^2 + (0+12)^2}$

$= \sqrt{81+144} = 15$ units

4. **(a);** y = mx + c
At x = 0, y = c

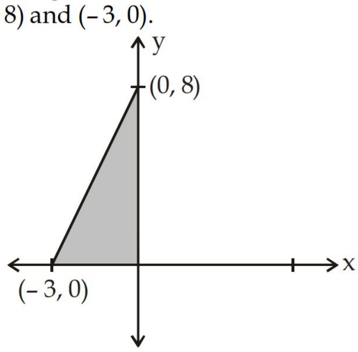
At y = 0, $x = \frac{-c}{m}$

The given line touches the coordinate axes at (0, c) and $\left(\frac{-c}{m}, 0\right)$.

Required length $= \sqrt{\left(0 - \left(-\frac{c}{m}\right)\right)^2 + (c-0)^2}$

$= \sqrt{\frac{c^2}{m^2} + c^2} = \frac{c}{m}\sqrt{1+m^2}$

5. **(b);** 8x – 3y = – 24
At x = 0, y = 8
At y = 0, x = – 3
So, the given line touches the coordinate axes at (0, 8) and (– 3, 0).



So, area of the triangle $= \frac{1}{2} \times \text{base} \times \text{height}$

$= \frac{1}{2} \times 3 \times 8 = 12$ sq. units

6. **(c);** 4x – 3y = – 4 (i)
4x + 3y = 20 (ii)
Adding (i) and (ii), we get
8x = 16 $\Rightarrow$ x = 2
Put x = 2 in (ii)
$\Rightarrow$ y = 4
So, the given lines intersect at (2, 4).
Now, the given lines intersect the x-axis at (– 1, 0). and (5, 0).
So, base of the triangle = 5 – (– 1) = 6
height = 4

Area $= \frac{1}{2} \times 6 \times 4 = 12$ sq. units

7. **(a);** $3x - 4y = 0$ ...(i)

$x = 4$ ...(ii)

The given lines intersect at (4, 3) and they intersect the x-axis at (0, 0) and (4, 0)

$\Rightarrow$ Base = 4 units

height = 3 units

$$\text{Area} = \frac{1}{2} \times 4 \times 3 = 6 \text{ sq. units}$$

8. **(b);** $3x + 4y = 24$ ...(i)

$x = 8$ ...(ii)

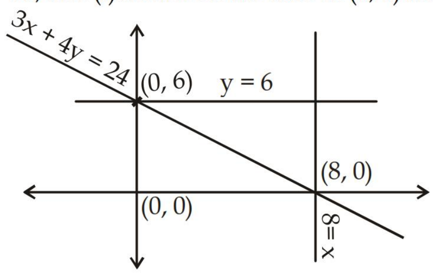
$y = 6$ ...(iii)

From (i):

At $x = 0,\ y = 6$

At $y = 0,\ x = 8$

So, line (i) intersects the axes at (0, 6) and (8, 0).



$$\text{Required area} = \frac{1}{2} \times 8 \times 6 = 24 \text{ sq. units}$$

9. **(d);** Checking each option one by one, in option (d):

$$\frac{a_1}{a_2} = 1,\quad \frac{b_1}{b_2} = -1$$

$$\text{i.e;}\quad \frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

So, system of equations given in option (d) has unique solution

10. **(b);** $2x - ky + 3 = 0$

$$y = \frac{2x+3}{k} = \frac{2}{k}x + \frac{3}{k}$$

$$\text{slope} = \frac{2}{k} = m,\quad 3x + 2y = 1$$

$$y = -\frac{3}{2}x + \frac{1}{2},$$

For the given lines to be parallel, slopes must be equal.

$$m_1 = \frac{2}{k} = -\frac{3}{2} = m_2 \quad \Rightarrow \quad k = -\frac{4}{3}$$

11. **(a);** The given equations are:

$5x + 2y - k = 0$ ...(i)

$10x + 4y - 3 = 0$ ...(ii)

$$\frac{a_1}{a_2} = \frac{5}{10} = \frac{1}{2},\quad \frac{b_1}{b_2} = \frac{2}{4} = \frac{1}{2},\quad \frac{c_1}{c_2} = \frac{-k}{-3} = \frac{k}{3}$$

For the given system to have infinitely many solutions:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\Rightarrow \frac{k}{3} = \frac{1}{2} \quad \Rightarrow \quad k = \frac{3}{2}$$

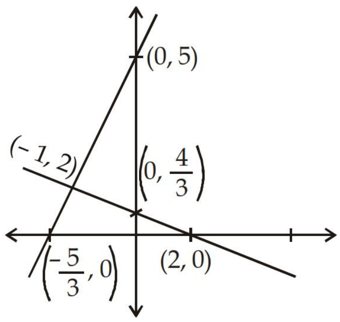
12. **(b);** Equation (i) $\rightarrow 2x + 3y = 4$

$$\text{At } x = 0,\ y = \frac{4}{3},\qquad \text{At } y = 0,\ x = 2$$

Equation (ii) $\rightarrow 3x - y = -5$

$$\text{At } x = 0,\ y = 5,\qquad \text{At } y = 0,\ x = -\frac{5}{3}$$

The graph of the above equations is as follows:



The given lines intersect at (– 1, 2).

$$\text{So, Required ratio} = \frac{\frac{1}{2}\left(2 - \left(-\frac{5}{3}\right)\right) \times 2}{\frac{1}{2}\left(5 - \frac{4}{3}\right) \times 1} = 2:1$$

$$= 2:1$$

13. **(d);** abscissa = positive, ordinate = negative so point will lie in fourth quadrant

14. **(b);** Distance between the given points

$$= \sqrt{(b-0)^2 + (0-a)^2} = \sqrt{a^2 + b^2}$$

15. **(a);**

16. **(c);** Write the equation in the form

$y = mx + c$

$3x + 7y + 8 = 0$

$$y = \frac{-3x - 8}{7} = -\frac{3}{7}x - \frac{8}{7}$$

$$\text{slope} = -\frac{3}{7}$$

17. **(b);**

18. **(a);** Points A and B both satisfy the equation given in option (a) only.

19. (c); Equation of line formed by the points P(2, 3) and R(6, 7) is:

$$y-3=\left(\frac{7-3}{6-2}\right)(x-2) \Rightarrow y-3=\frac{4}{4}(x-2)$$

$\Rightarrow$ y = x + 1

Now, Q should satisfy the above equation for the three points to be collinear

$\Rightarrow$ a = 5 + 1 = 6

20. (a); Let the ratio be k : 1

x– coordinate of a point lying on y-axis must be 0.

$$\frac{7k-3}{k+1}=0 \Rightarrow k=\frac{3}{7}$$

So, required ratio = 3 : 7

21. (c); x + 3y – 8 = 0

$$\Rightarrow y=\frac{-x+8}{3}=-\frac{1}{3}x+\frac{8}{3}$$

$$\therefore \text{ slope}=-\frac{1}{3}$$

ax + 12y + 5 = 0

$$y=-\frac{ax}{12}-\frac{5}{12}$$

$$\text{Slope}=-\frac{a}{12}$$

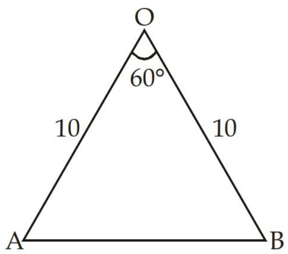
For the lines to be parallel, slopes must be equal

$$\Rightarrow -\frac{1}{3}=-\frac{a}{12} \Rightarrow a=4$$

22. (a); The required coordinates are:

$$\frac{2\times\frac{11}{2}+3\times 3}{2+3}, \frac{2\times\frac{21}{2}+3\times(-2)}{2+3}=(4, 3)$$

23. (b);

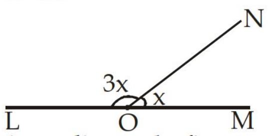


OA and OB are two legs of the compass.

Clearly, the above triangle is an equilateral triangle.

So, distance between end points of the legs is: 10 cm

24. (a);



According to the figure:

$4x = 180° \Rightarrow x = 45° = \angle MON$

25. (b); Let $A=(4,3)$, $B=(7,-1)$ and $C=(9,3)$

then $AB=\sqrt{(7-4)^2+(-1-3)^2}=\sqrt{9+16}=5$

$$BC=\sqrt{(9-7)^2+(3+1)^2}=\sqrt{20}=2\sqrt{5}$$

$$CA=\sqrt{(4-9)^2+(3-3)^2}=\sqrt{5^2}=5$$

Therefore it is an isosceles triangle.

26. (b); The required coordinates are

$$\left(\frac{mx_2-nx_1}{m-n}, \frac{my_2-ny_1}{m-n}\right)$$

$(x_1, y_1) = (2, 4)$ $(x_2, y_2) = (6, 8)$

m : n = 5 : 3

$$x=\frac{5\times 6-3\times 2}{5-3}=\frac{24}{2}=12$$

$$y=\frac{5\times 8-3\times 4}{5-3}=\frac{28}{2}=14$$

The required coordinates are (12, 14)

27. (a); Coordinate of centroid of ΔABC be (x, y) then

$$(x,y)=\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$

$$=\left(\frac{3+2-2}{3}, \frac{1+3+2}{3}\right)=(1,2)$$

28. (c); $AB=\sqrt{(1+2)^2+1^2}=\sqrt{10}$

$$BC=\sqrt{(4-1)^2+3^2}=\sqrt{9+9}=\sqrt{18}$$

$$CD=\sqrt{(1-4)^2+(2-3)^2}=\sqrt{9+1}=\sqrt{10}$$

$$DA=\sqrt{(-2-1)^2+(2+1)^2}=\sqrt{9+9}=\sqrt{18}$$

The opposite sides are equal, Hence, it is a parallelogram

29. (a); Slope of line $=\frac{y_2-y_1}{x_2-x_1}=\frac{7-5}{9-7}=1$

30. (a); m_1 = Slope of line AB $=\frac{4+2}{3-1}=\frac{6}{2}=3$

m_2 = Slope of line BC $=\frac{7-4}{4-3}=3$

$m_1 = m_2$

AB is parallel to BC and B is common. Therefore, it is a straight line.

Moderate

1. **(a);** Slope = $\frac{7-3}{-4-2}=\frac{4}{-6}=\frac{-2}{3}$

2. **(c);** Area of ΔPQR when the vertices are $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ is given by the formula;

$$\text{Area} = \frac{1}{2}x_1(y_2-y_3)+x_2(y_3-y_1)$$
$$+x_3(y_1-y_2)$$

So, area of ΔPQR =

$$\frac{1}{2}[4(8-(-4)+(-3)(-4-5)+3(5-8)]$$

$$=\frac{1}{2}|48+27+(-9)|$$

$$=\frac{1}{2}\times 66 = 33 \text{ Sq. units.}$$

3. **(d);** Co-ordinates of the centroid of ΔPQR are:

$$\left(\frac{-2+9+8}{3},\frac{0-3+3}{3}\right)\equiv(5,0)$$

4. **(a);** Slope of the joining the points $(\sqrt{3},1)$ and $(\sqrt{15},\sqrt{5})$ = $\tan\theta$

$$=\frac{\sqrt{5}-1}{\sqrt{15}-\sqrt{3}}=\frac{(\sqrt{5}-1)}{\sqrt{3}(\sqrt{5}-1)}=\frac{1}{\sqrt{3}}$$

$\Rightarrow$ The required angle = 30°

5. **(d);** According to the question:
$(x-a)^2+y^2+(x+a)^2+y^2=2b^2$
$\Rightarrow\ x^2+a^2=b^2-y^2$

6. **(b);** Let the co-ordinate of R be (x, y).
According to the question:

$$\frac{-1+5+x}{3}=4 \quad\Rightarrow\quad x=8$$

$$\frac{0-2+y}{3}=0 \quad\Rightarrow\quad y=2$$

So, Co-ordiantes of R is
(8, 2)

7. **(b);** Let the ratio be K : 1
on the x-axis, y-coordinate is zero

$$\Rightarrow\quad \frac{4k-5}{k+1}=0 \quad\Rightarrow\quad K=\frac{5}{4}$$

So, the required ratio is 5 : 4.

8. **(d);** For the lines to be perpendicular to each other, product of slopes must be equal to –1

$$\text{Slope of line PQ} = m_1 = \frac{-7-5}{0+2}=-6$$

$$\text{Slope of line AB} = m_2 = \frac{a+2}{8+4}=\frac{a+2}{12}$$

$m_1m_2 = -1$

$$\Rightarrow\quad -6\times\left(\frac{a+2}{12}\right)=-1 \quad\Rightarrow\quad a=0$$

9. **(a);** The given lines can be represented as:

$$y=\frac{\sqrt{12}\,x}{2}+\frac{9}{2}$$

$$\text{and } y=\frac{x}{\sqrt{3}}-\frac{7}{\sqrt{3}}$$

So, the slopes are:

$$m_1=\frac{\sqrt{12}}{2},\quad m_2=\frac{1}{\sqrt{3}}$$

The tangent of the angle between the given lines:

$$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|=\frac{1}{\sqrt{3}}$$

$\Rightarrow\quad \theta = 30°.$

10. **(d);** Given, straight lines:
$2x+3y=5$ ... (i)
$3x-y=13$ ... (ii)
Solving the above two equations, we get the intersection point:
(4, – 1).
From equation (i):

$$\text{at } y=0,\ x=\frac{5}{2}.$$

From the equation (ii);

$$\text{At } y=0,\ x=\frac{13}{3}$$

So, the three vertices of the triangle are :

$$(4,-1),\left(\frac{5}{2},0\right),\left(\frac{13}{3},0\right)$$

$$\text{Area}=\frac{1}{2}\left|4(0-0)+\frac{5}{2}(0+1)+\frac{13}{3}(-1-0)\right|$$

$$=\frac{11}{12}\text{ sq. units.}$$

11. (d); Given straight lines:

$4x - y = 4$ ... (i)

$3x + 2y = 14$ ... (ii)

Solving the above two equations we get the intersection point:

(2, 4)

On the y-axis, x-coordiate is zero.

From (i): At x = 0, y = – 4

From (ii): At x = 0, y = 7

So, the vertices of the triangle are:

(2, 4), (0, – 4) and (0, 7)

$$\text{Area} = \frac{1}{2}\left|2(-4-7)+0+0\right| = 11 \text{ sq. units.}$$

12. (b); Given straight lines:

$2x - 3y = -6$ ... (i)

$2x + 3y = 18$ ... (ii)

$y = 1$ ... (iii)

Form (i) and (iii):.

$2x - 3 = -6$

$$\Rightarrow \quad x = \frac{-3}{2}$$

From (ii) and (iii).

$2x + 3 = 18$

$$\Rightarrow \quad x = \frac{15}{2}$$

From (i) and (ii):

x = 3

y = 4

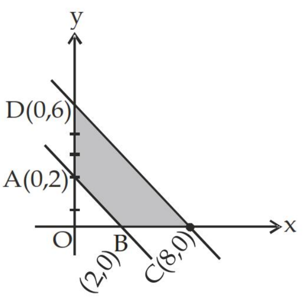
So, the vertices of the triangle are:

$\left(\frac{-3}{2}, 1\right), \left(\frac{15}{2}, 1\right)$ and (3, 4).

$$\text{Area} = \frac{1}{2}\left|\frac{-3}{2}(1-4)+\frac{15}{2}(4-1)+3(1-1)\right|$$

$$= \frac{27}{2} \text{ sq. units.}$$

13. (a);



$$\text{Area of } \Delta DOC = \frac{1}{2} \times 8 \times 6 = 24 \text{ sq. units.}$$

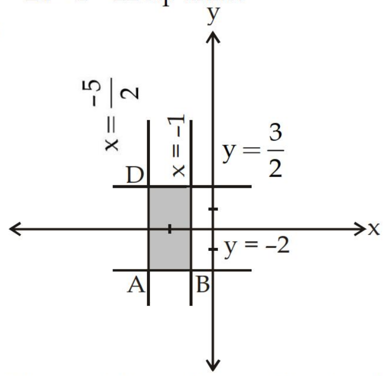
$$\text{Area of } \Delta AOB = \frac{1}{2} \times 2 \times 2 = 2 \text{ sq. units.}$$

Area of Quad. ABCD =

Area of ΔDOC – Area of ΔAOB

= 24 – 2 = 22 sq. units.

14. (b);

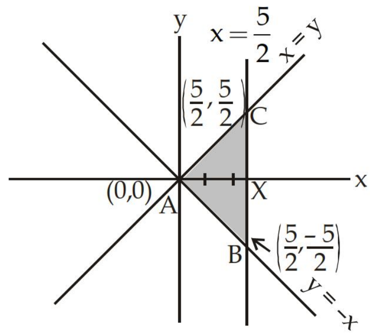


The quadrilateral formed by the given lines is a rectangle.

Area = L × B

$$= \frac{7}{2} \times \frac{3}{2} = \frac{21}{4} \text{ sq. units}$$

15. (c);



$$\text{Length of base BC of } \Delta ABC = \frac{5}{2}+\frac{5}{2} = 5 \text{ units}$$

$$\text{Height} = AX = \frac{5}{2} \text{ units.}$$

$$\text{Area} = \frac{1}{2} \times 5 \times \frac{5}{2} = \frac{25}{4} \text{ sq. units.}$$

16. (a); Given equations:

$2x + 3y - 7 = 0$ ... (i)

$2ax + (a + b)y - 28 = 0$ ... (ii)

For infinitely many solutions:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

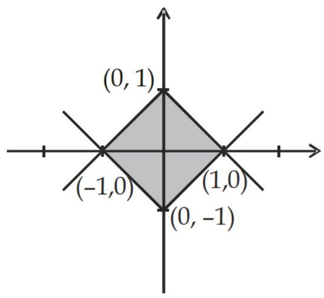
$$\Rightarrow \quad \frac{2}{2a} = \frac{3}{a+b} = \frac{-7}{-28}$$

$$\Rightarrow \quad \frac{1}{a} = \frac{7}{28} = \frac{1}{4} \Rightarrow a = 4$$

$$\text{and, } \frac{1}{a} = \frac{3}{a+b} \Rightarrow b = 2a = 8$$

17. (a); Given equations: $y = |x| - 1$

$y = 1 - |x|$



Each side of the figure formed by the given equations = $\sqrt{2}$ units

Area = $\sqrt{2} \times \sqrt{2} = 2$ sq. units.

18. (b); Fig

Given $AC^2 = AB \times CB$

$\Rightarrow \quad x^2 = 2 \times (2 - x)$

$x^2 = 4 - 2x$

$x^2 + 2x - 4 = 0$

$ax^2 + bx + c = 0$

$$x = \frac{-2 \pm \sqrt{4+16}}{2 \times 1} = 1 \pm \sqrt{5}$$

Now, $BC = 2 - (-1 \pm \sqrt{5}) = 3 - \sqrt{5}$

$\left(3 + \sqrt{5} > AB, \text{ which is not possible}\right)$

19. (c); Sum of all angle exculded angle of triangle are = $360° + 360° + 360° - 180°$

$= 1080° - 180° = 900°$

20. (a); Slope of AB = $\dfrac{5+3}{2-x} = \dfrac{8}{2-x} = \tan 135°$

$\tan(180° - 45°) = \dfrac{8}{2-x}$

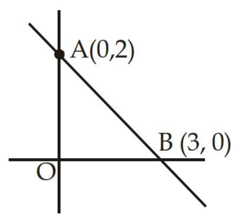
$-\tan 45° = \dfrac{8}{2-x} \quad \Rightarrow \quad -1 = \dfrac{8}{2-x}$

$\Rightarrow \quad 2 - x = -8$

$x = 8 + 2$

$x = 10$

21. (b); $2x + 3y = 6$



$$\frac{2x}{6} + \frac{3y}{6} = 1$$

$$\frac{x}{3} + \frac{y}{2} = 1$$

Therefore It intercept at x-axis at 3 and intercept at y-axis = 2, Area of $\Delta OAB = \dfrac{1}{2} \times 3 \times 2 = 3$ sq. units

22. (a); $x + 2y = 9$

$kx + 4y = -5$

For equation of line to be parallel $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2}$

$\dfrac{1}{k} = \dfrac{2}{4} \quad k = 2$

23. (d); Slope of the line made by joining the points A (1, 1) and B(3, 5) = $\dfrac{5-1}{3-1} = \dfrac{4}{2} = 2$ and mid point of AB = $\left(\dfrac{1+3}{2}, \dfrac{1+5}{2}\right)$ = (2, 3)

Now, slope of the line perpendicular to the line AB = $\dfrac{-1}{2}$

$\therefore$ Required equation of perpendicular bisector

$(y - 3) = -\dfrac{1}{2}(x - 2)$

$x + 2y - 8 = 0$

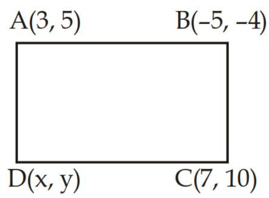
24. (b); Let the fourth vertex be (x, y)

Diagonal of parallelogram intersect at mid point

Therefore

midpoint of AC = Mid-point of BD

$$\left(\frac{3+7}{2}, \frac{5+10}{2}\right) = \left(\frac{-5+x}{2}, \frac{-4+y}{2}\right)$$



$\Rightarrow \quad 5 = \dfrac{-5+x}{2}$

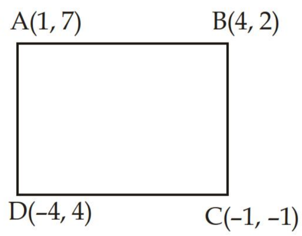
$x = 10 + 5 = 15$

$\dfrac{5+10}{2} = \dfrac{-4+y}{2}$

$y = 19$

Fourth vertex will be (15, 19)

25. (b);



$AB = \sqrt{(4-1)^2 + (2-7)^2} = \sqrt{9+25} = \sqrt{34}$

$BC = \sqrt{(-1-4)^2 + (-1-2)^2} = \sqrt{5^2+3^2} = \sqrt{34}$

$CD = \sqrt{(-4+1)^2 + (+4+1)^2} = \sqrt{3^2+5^2} = \sqrt{34}$

$DA = \sqrt{(1+4)^2 + (7-4)^2} = \sqrt{5^2+3^2} = \sqrt{34}$

$AC = \sqrt{(-1-1)^2 + (-1-7)^2} = \sqrt{2^2+8^2} = \sqrt{68}$

$BD = \sqrt{(-4-4)^2 + (4-2)^2} = \sqrt{8^2+2^2} = \sqrt{68}$

Given, AB = BC = CD = DA and AC = BD

Therefore it is a square.

26. **(b);** Since, P, Q and R are collinear

Area of $\Delta PQR = 0$

$\frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] = 0$

Given $x_1 = a$, $x_2 = 0$

$x_3 = 1$, $y_1 = 0$, $y_2 = b$ and $y_3 = 1$

$\frac{1}{2}[a(b-1) + 0(1-0) + 1(0-b)] = 0$

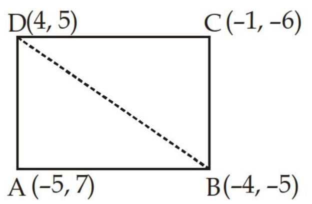
$ab - a - b = 0$

$ab = a + b$

$1 = \frac{1}{a} + \frac{1}{b}$

Difficult

1. **(a);**



Join BD

Given $x_1 = -5$, $x_2 = -4$, $x_3 = 4$

$y_1 = 7$, $y_2 = -5$ and $y_3 = 5$

Area of $\Delta ABD = \frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1)$

$+ x_3(y_1 - y_2)]$

$= \frac{1}{2}[-5(-5-5) + (-4)(5-7) + 4(7+5)]$

$= \frac{1}{2}[50 + 8 + 48] = \frac{106}{2} = 53$ sq. units

Now In ΔBCD

Given $x_1 = -4_1$, $x_2 = -1$, $x_3 = 4$, $y_1 = -5$, $y_2 = -6$, $y_3 = 5$

Area of $\Delta BCD = \frac{1}{2}[(-4)(-6-5) - 1(5+5)$

$+ 4(-5+6)]$

$= \frac{1}{2}(44 - 10 + 4) = \frac{1}{2} \times 38 = 19$ sq. units.

Area of quadrilateral ABCD = 19 + 53 = 72 sq. units

2. **(d);** $x + 2y = 3$ ... (i)

$5x + y = -12$ ... (ii)

On solving (i) and (ii)

We get $9x = -27$

$x = \frac{-27}{9} = -3$

$2y = 6$, $y = 3$

Coordinate of P(–3, 3)

Similary Q = (1, 1)1 R(1, –1) and S(–2, –2)

Now M_1 = solpe of PR $= \frac{-1-3}{1+3} = \frac{-4}{4} = -1$

m_2 = solpe of QS $= \frac{-2-1}{-2-1} = 1$

$m_1 m_2 = -1$

the required angle is 90°

3. **(a);** P_1 = length of perpendicular from (0, 0) on $x \sec\theta + y \operatorname{cosec}\theta$

$P_1 = \frac{a}{\sqrt{\sec^2\theta + \operatorname{Cosec}^2\theta}} = \frac{a}{\sqrt{\frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta}}}$

$= a \sin\theta \cos\theta$

or $2p_1 = a\,(2\sin\theta.\cos\theta)$

$\Rightarrow 2p_1 = a \sin 2\theta$

Similarly, $P_2 = \frac{a\cos 2\theta}{\sqrt{\cos^2\theta + \sin^2\theta}} = a\cos 2\theta$

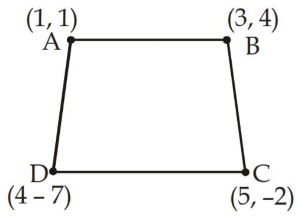
$4p_1^2 = a^2 \sin^2 2\theta$

$P_2^{\,2} = a^2 \cos^2 2\theta$

$4P_1^{\,2} + P_2^{\,2} = a^2 \sin^2 2\theta + a^2 \cos^2 2\theta$

$= a^2(\sin^2 2\theta + \cos^2 2\theta) = a^2$

4. **(a);**



Hence $x_1 = 1, x_2 = 3,$

$x_3 = 4$

$y_1 = 1, y_2 = 4, \quad y_3 = -7$

Area of $\Delta ABD = \frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1)$

$+ x_3(y_1 - y_2)]$

$= \frac{1}{2}[1(11) + 3(-8) + 4(-3)]$

$= \frac{1}{2}(11 - 24 - 12) = \frac{1}{2} \times 25 = \frac{25}{2}$ sq, unit

$x_1 = 3, x_2 = 5, x_3 = 4, y_1 = 4, y_2 = -2, y_3 = -7$

Area of $\Delta BCD = \frac{1}{2}[3(5) + 5(-11) + 4(6)]$

$= \frac{1}{2}[15 - 55 + 24] = \frac{1}{2}[16] = 8$ sq.units

Area of avadrilateral ABCD

= or (ΔABD) + ar (ΔBCD)

$= \frac{25}{2} + 8 = 20.5$ sq.units

5. **(b);** Slope of line joining (1, 2) and (2, –2) = $\frac{-4}{1} = -4$

Slope of line joining (8, 2) and (4, p)

$= \frac{-(p-2)}{4} = \left(\frac{2-p}{4}\right)$

For perpendicular

$m_1 \cdot m_2 = -1$

$-4 \times \frac{2-p}{4} = -1$

$2 - p = 1$

$p = 1$

6. **(a);** Point of intersections

$\frac{bx + ay}{ab} = 1, bx + ay = ab$ ... (i)

$\frac{ax + by}{ab} = 1, ax + by = ab$ ... (ii)

Multiply eq (i) by a and eq (ii) by b

$abx + a^2y = a^2b$

$abx + b^2y = ab^2$

$(a^2 - b^2)\, y = a^2b - ab^2$

$y = \frac{ab(a-b)}{(a-b)(a+b)} = \frac{ab}{a+b}$

Putting this value in eq (i)

$bx + a\left(\frac{ab}{a+b}\right) = ab$

$bx = \frac{ab^2}{a+b}, x = \frac{ab}{a+b}$

Therefore equation of line is $y = x$

7. **(c);** $2x + y = 4$ ... (i)

$x - y + 1 = 0$ ... (ii)

Solving eq^n (i) and (ii)

$x = 1$ and $y = 2$

The point of intersection of (i) and (ii) is (1, 2)

Again on solving $2x - y - 1 = 0$ ... (iii)

$x + y - 8 = 0$ ... (iv)

$x = 3, y = 5$

The point of intersection of (iii) and (iv) is (3, 5).

The required equation of straight line joing the point of intersection

$y - 2 = \frac{5-2}{3-1}(x - 1)$

$y - 2 = \frac{3}{2}(x - 1), \quad 2y - 4 = 3x - 3$

$3x - 2y + 1 = 0$

8. **(d);** Equation of line passing through point (4, 5)

$y - 5 = m(x - 4)$ ... (i)

where m is the slope of the line.

Now the given line is $2x - y + 7 = 0$

$y = 2x + 7$ ... (ii)

If m_1 be the slope of eq^n (ii)

$m_1 = 2$

$$\tan 45° = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{2 - m}{1 + 2m}$$

$1 + 2m = 2 - m$

$$3m = 1, \ m = \frac{1}{3}$$

$$y - 5 = \frac{1}{3}(x - 4)$$

$3y - 15 = x - 4, \quad x - 3y + 11 = 0$

9. (b); The equation of line passing through the point of intersection of the lines $3x - 4y + 1 = 0$ and $5x + y - 1 = 0$ is $(3x - 4y + 1) + k(5x + y - 1) = 0$

For intercept of this line with x axis, $y = 0$

$$3x + 1 + k(5x - 1) = 0, \quad x = \frac{k - 1}{5k + 3}$$

Intercept on y-axis, $x = 0$

$$-4y + 1 + k(y - 1) = 0, \quad y = \frac{k - 1}{k - 4}$$

Since the intercepts on the axes are equal.

$$\frac{k-1}{5k+3} = \frac{k-1}{k-4}, \ k = 1 \quad \text{or} \quad K = \frac{-7}{4}$$

For $k = 1$ equation becomes $8x - 3y = 0$

but it does not cut equal intercepts on coordinate axis

$k = \frac{-7}{4}$ and from (i) we get

$3x - 4y - 1 - 7(5x + y - 1) = 0$

$23x + 23y = 11$

which is required equation of the line.

10. (c); Let A(1, 2), B(2, 3) and C(4, 3) be the vertices of ΔABC

$$m_1 = \text{slope of BC} = \frac{3 - 3}{4 - 2} = 0$$

$$m_2 = \text{slope of line AC} = \frac{3 - 2}{4 - 1} = \frac{1}{3}$$

$$m_3 = \text{slope of line AB} = \frac{3 - 2}{2 - 1} = 1$$

The equation of line perpendicular from B(2, 3) on AC whose slope is -3 is

$y - 3 = -3(x - 2)$

$y - 3 = -3x + 6$

$3x + y - 9 = 0$ ... (i)

The perpendicular from A (1, 2) to BC is parallel to y axis. Equation of line perpendicular from A (1, 2) on BC is $x = 1$ ... (ii)

The orthocentre is the point of intersection of two lines (i) and (ii)

we get $= 3 \times 1 + y = 9$

$y = 6$

coordinate of orthocentre is (1, 6)

11. (d); Let x_1 and y_1 be the third vertex (x_1, y_1)

then $y_1 = x_1 + 3$ ... (i)

The area of triangle formed by point (2, 1) (3, -2) and (x_1, y_1)

$$= \frac{1}{2}|-7 + 3y_1 + 2x_1 + x_1 - 2y_1|$$

$$= \frac{1}{2}|3x_1 + y_1 - 7|$$

Area of triangle is 5

$$5 = \frac{1}{2}|3x_1 + y_1 - 7|$$

$10 = 3x_1 + y_1 - 7, \ 10 = -3x_1 - y_1 + 7$

$3x_1 + y_1 = 17, \quad 3x_1 - y_1 = -3$ ... (ii)

On solving (i) and (ii)

$$x_1 = \frac{7}{2} \text{ and } y_1 = \frac{13}{2}, \text{ or } x_1 = \frac{-3}{2}, y_1 = \frac{3}{2}$$

12. (a); Equation of any line L perpendicular to $5x - y = 1$ is $x + 5y = k$ ... (i)

where k is an arbitrary constant.

$$\text{when } x = 0, \quad y = \frac{k}{5} \quad \left(0, \frac{k}{5}\right)$$

when $y = 0, \quad x = k \qquad (k, 0)$

$$\text{Area of triangle} = \frac{1}{2}(x_1 y_2 - x_2 y_1)$$

$$= \frac{1}{2}\left(\frac{k^2}{5} - 0\right) = \frac{k^2}{10}$$

But Area given is 5.

$$5 = \frac{k^2}{10}, K = \pm 5\sqrt{2}$$

Equation of line $x + 5y = \pm 5\sqrt{2}$

13. (a); $x + y - 8 = 0$

$3x - 2y + 1 = 0$

Solving above equations, the point of intersection is (3, 5).

Now equation of line parallel to the line $x - y + 1 = 0$ is

$x - y + c = 0$ ... (i)

$3 - 5 + c = 0, c = 2$

From eq^n (i)

$x - y + 2 = 0$

14. (b); $x + 2y - 3 = 0$

$4x - y + 7 = 0$

Point of intersection on solving above equations

$$x = -\frac{11}{9}, \quad y = \frac{19}{9}$$

Equation of line

$$y - \frac{19}{9} = 1.\left(x + \frac{11}{9}\right)$$

$9y - 19 = 9x + 11$

$9x - 9y = -30 \quad \Rightarrow \quad 3y - 3x = 10$

15. (d); $\frac{x}{a} - \frac{y}{b} = 1$ ... (i)

It passes through the point (8, 6)

$$\frac{8}{a} - \frac{6}{b} = 1$$

The line meet x – axis at $y = 0$ from (i) $x = a$ coordinate of point A is (a, 0)

Similarly, coordinate of point B (0, – b)

Area given = 12

$$\frac{1}{2}ab = 12$$

$$ab = 24, \quad b = \frac{24}{a}$$

$$\frac{8}{a} - \frac{6}{\frac{24}{a}} = 1 \quad a = 4 \text{ or } -8$$

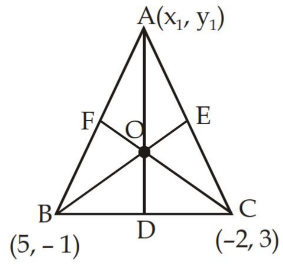
and $b = 6$ or -3

Hence from (i) the equation of straight line are

$$\frac{x}{4} - \frac{y}{6} = 1 \quad \text{and} \quad \frac{x}{-8} - \frac{y}{-3} = 1$$

$3x - 2y = 12$ and $3x - 8y + 24 = 0$.

16. (c);



Let AD, BE and CF be the perpendicular from the vertices on the opposite side BC, AC and BA respectively.

Orthocentre at origin

O(0, 0)

Since AD or OA is perpendicular to BC

Slope OA x slope BC = –1

$$\frac{y_1 - 0}{x_1 - 0} \times \frac{3 - (-1)}{-2 - 5} = -1$$

$$y_1 = \frac{7x_1}{4} \quad \text{... (i)}$$

Again $OB \perp CA$

$$\frac{-1 - 0}{5 - 0} \times \frac{y_1 - 3}{x_1 + 2} = -1$$

$5x_1 + 10 = y_1 - 3$

$x_1 = -4$

From (i)

$$y_1 = \frac{7x_1}{4} = \frac{7 \times (-4)}{4} = -7$$

Required coordinates of vertex A are (– 4, – 7)

17. (d); The equation of the lines may be written as $3x + 4y + 2 = 0$ and $-5x + 12y + 6 = 0$ in which both the constant terms 2 and 6 are positive.

The equation of the bisector of the angle in which the origin lies $\frac{3x + 4y + 2}{\sqrt{3^2 + 4^2}} = \frac{-5x + 12y + 6}{\sqrt{(-5)^2 + 12^2}}$

$16x - 2y - 1 = 0$

The equation of other bisector in

$$\frac{3x + 4y + 2}{\sqrt{3^2 + 4^2}} = \frac{(+5x - 12y - 6}{\sqrt{5^2 + 12^2}}$$

$x + 8y + 4 = 0$

18. **(d);** Since the line passes through A(a, 0) and B(0, b) it makes intercepts a and b on the axes of 'x' and 'y'. Let the equation of the line be

$$\frac{x}{a}+\frac{y}{b}=1 \quad \text{... (i)}$$

By the condition

length of segment = 13

$a.b = 60$ ... (ii)

From (i) $\sqrt{a^2+b^2}=13$

$a^2 + b^2 = 169$

$a + b = \pm 17$

$(a - b)^2 = (a + b)^2 - 4ab = 289 - 240 = 49$

we get a = 12, b = 5 and a = – 12, b = – 5

Required equations

$$\frac{x}{12}+\frac{y}{5}=1 \qquad \frac{x}{-12}+\frac{y}{-5}=1$$

$5x + 12y = 60 \quad 5x + 12y = -60$

19. **(d);** Intercept from $\frac{x}{a}+\frac{y}{b}=1$

a + b = 14 or b = 14 – a

line passes through the point (3, 4)

$$\frac{3}{a}+\frac{4}{b}=1, \frac{3}{a}+\frac{4}{14-a}=1$$

a = 6, 7

If a = 6, then b = 8

a = 7, then b = 7

Required equations $\frac{x}{6}+\frac{y}{8}=1$ and $\frac{x}{7}+\frac{y}{7}=1$

$4x + 3y = 24, x + y = 7$

20. **(c);** Let the vertices be A(4, –1) B(6, 0), C (7, 2), D(5, 1) then the coordinate of midpoint of AC are

$$\left(\frac{4+7}{2},\frac{-1+2}{2}\right)=\left(\frac{11}{2},\frac{1}{2}\right)$$

Coordinate of mid point of BD are

$$\left(\frac{6+5}{2},\frac{0+1}{2}\right)=\left(\frac{11}{2},\frac{1}{2}\right)$$

Since AC and BD have same mid point. Hence ABCD is a parallelogram.

Now, AB = $\sqrt{5}$, BC = $\sqrt{5}$, AB = BC

We have AC = $3\sqrt{2}$ BD = $\sqrt{2}$, AC $\neq$ BD

Therefore it is a rhombus.

Previous Year (Memory Based)

1. **(b);** At y-axis, x = 0

$2x - 3y = 6$

$2(0) - 3y = 6, y = \frac{-6}{3} = -2$

coordinates of point are (0, –2)

2. **(c);** The line passing through the origin must satisfy the equation for (0, 0).

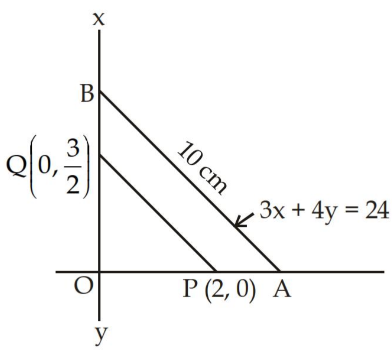
On putting x = 0 in

$2x + 3y = 0$

$2(0) + 3y = 0 \quad y = 0$

$2x + 3y = 0$ satisfy the coordinates (0, 0).

3. **(b);**



$3x + 4y = 24$

$$\frac{x}{8}+\frac{y}{6}=1$$

X-intercept, OA = 8

Y-intercept, OB = 6

Slope of line PQ = $\frac{3/2-0}{0-2}=\frac{-3}{4}$

Slope of line AB, 3x + 4y = 24 is $\frac{-3}{4}$

Therefore PQ | | AB

$\Delta OPQ \sim \Delta OAB$

$$\frac{OP}{OA}=\frac{PQ}{AB} \quad \frac{2}{8}=\frac{PQ}{10}$$

$$PQ=\frac{2\times 10}{8}=2.5 \text{ cm}$$

4. **(a);** Convert this equation in form $\frac{x}{a}+\frac{y}{b}=1.$

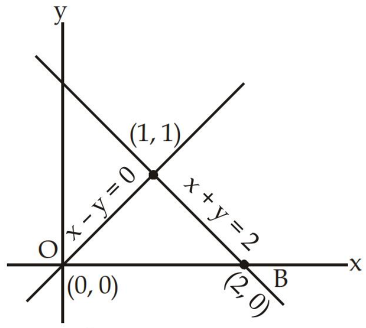
$9x - 12y = 108$

$$\frac{9x}{108}+\frac{(-12)y}{108}=\frac{108}{108} \quad \frac{x}{12}+\frac{y}{(-9)}=1$$

Length of intercept = $\sqrt{a^2+b^2}$

$= \sqrt{12^2+9^2}=\sqrt{144+81}=15$ units

5. (a);



$x - y = 0$ ... (i)

$x + y = 2$ ... (ii)

On solving (i) and (ii)

$2x = 2, x = 1, y = 1$

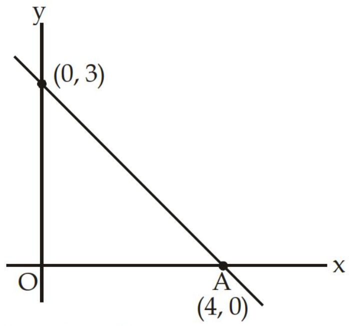
A (1, 1), B (2, 0) and C(0, 0)

$$\text{are }(\Delta AOB) = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \frac{1}{2}\times[1(0)+2(0-1)+0]$$

$$= \frac{1}{2}\times 2 = 1 \text{ sq. unit.}$$

6. (a);



$3x + 4y = 12$

$$\frac{3x}{12}+\frac{4y}{12}=\frac{12}{12}$$

$$\frac{x}{4}+\frac{y}{3}=\frac{12}{12}$$

$$\text{or }(\Delta AOB) = \frac{1}{2}\times OA\times OB$$

$$= \frac{1}{2}\times 4\times 3 = 6 \text{ sq. units}$$

7. (d); At x-axis, y coordinate is always zero.

$7x - 3y = 2$

$$\frac{x}{\frac{2}{7}}+\frac{y}{\left(-\frac{2}{3}\right)}=1$$

x intercept is $\frac{2}{7}$

8. (c); $2x + y = 6$ and $x = 2$

putting $x = 2$,

$2(2) + y = 6, y = 2$

Required Point = (2, 2)

9. (c); On putting $y = 0$ in equation, we get x intercept.

$5x + 6y = 30, 5x + 6(0) = 30$

$x = \frac{30}{5}, \; x = 6$ units.

10. (b); $2x + y = 5$ ... (i)

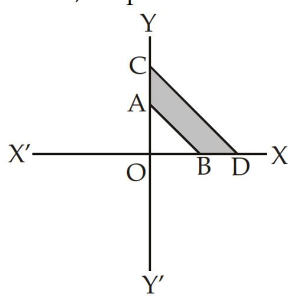
$x + 2y = 4$ ... (ii)

On solving (i) and (ii)

$y = 1$ and $x = 2$

Hence, the point of intersection is (2, 1)

11. (c);



$x = 0, y = 0, x + y = 1$

and $2x + 3y = 6$

On solving these equation

We get

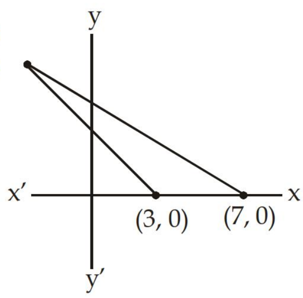
A(0, 1), B(1, 0) C(0, 2) and D(3, 0)

Area of region ABCD = ar (ΔOCD) – ar (ΔOAB)

$$= \frac{1}{2}\times 3\times 2-\frac{1}{2}\times 1\times 1$$

$$3-\frac{1}{2}=2\frac{1}{2} \text{ sq. units}$$

12. (a); $\left(\frac{-21}{13},\frac{80}{13}\right)$



$5x + 7y = 35$ ... (i)

$4x + 3y = 12$ ... (ii)

On solving 4 × (i) – 5 × (ii)

$28y - 15y = 140 - 60$

$13y = 80$

$y = \frac{80}{13}$

On solving equation (i) $x = \frac{-21}{13}$

Area of triangle $= \frac{1}{2} \times (7-3) \times \frac{80}{13}$

$= \frac{1}{2} \times 4 \times \frac{80}{13} = \frac{160}{13}$ sq. units .

13. (c); System of equation

$x + 2y - 5 = 0$ and $3x + ky + 15 = 0$

does not have solution when $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

$\frac{1}{3} = \frac{2}{k}$ and $\frac{2}{k} \neq \frac{-5}{15}$

$k = 6$ and $k \neq -6$

Hence $k = 6$

14. (d); $\because$ (a, b) lies on the straight lines $x = 3y - 7$

$a = 3b - 7$ ... (i)

$(a + 3, b + k)$ lies on the straight line $x = 3y - 7$

$a + 3 = 3(b + k) - 7$

$a + 3 = 3b + 3k - 7$ ... (ii)

subtracting (i) from (ii)

$3 = 3k, k = 1$

15. (d); $8x + 15y = 120$

$\frac{x}{15} + \frac{y}{8} = 1$

length intercpted b/w x-axis and y-axis

$= \sqrt{15^2 + 8^2} = \sqrt{289} = 17$

16. (a); Solving $2x + y = 4$ and $x - y + 1 = 0$, we get $x = 1$ and $y = 2$, so vertex (1, 2) and height is 2 units

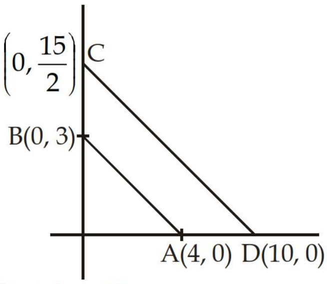
17. (a); $5x - 4y = 20$

Put $y = 0$

$5x - 4(0) = 20$

$5x = 20, x = \frac{20}{5} = 4$ units

18. (a);



$3x + 4y = 12$

$\frac{3x}{12} + \frac{4y}{12} = \frac{12}{12}, \quad \frac{x}{4} + \frac{y}{3} = 1$

$6x + 8y = 60$

$\frac{6x}{60} + \frac{8y}{60} = \frac{60}{60}, \quad \frac{x}{10} + \frac{y}{15/2} = 1$

Area of trapezium ABCD = ar (ΔOCD) – or (ΔOAB)

$= \frac{1}{2} \times 10 \times \frac{15}{2} - \frac{1}{2} \times 3 \times 4 = 37.5 - 6$

$= 31.5$ sq. units

19. (b); $2x - 3y + 6 = 0$

$2x - 3y = -6$

$\frac{x}{(-3)} + \frac{y}{2} = 1$

Coordinate with axis are (–3, 0) and (0, 2)

Area $= \frac{1}{2} \times 3 \times 2 = 3$ sq. units

20. (c); Let (x, 0) be the centre

radius $= \sqrt{(x-4)^2 + 5^2}$

$5^2 = (x-4)^2 + 5^2$

$x = 4$

Therefore centre of circle is (4, 0)

21. (d); Let A(0, 0), B(3, 3) and C(k, 0) be the given point From the choices when $k = 6$

$AB = \sqrt{3^2 + 3^2} = \sqrt{9+9} = \sqrt{18}$ units .

$BC = \sqrt{(3-6)^2 + (3-0)^2} = \sqrt{18}$ units

$CA = \sqrt{(6-0)^2 + (0-0)^2} = \sqrt{36}$ units

$AB^2 + BC^2 = CA^2$

$\angle B = 90°$ $AB = BC$

Therefore when $k = 6$ ΔABC is a right angled isoceles triangle.

22. (d); Equation of line passing through (2, 2) and (5, 5) is $x = y$

This passes through (a, 4) when $a = 4$

23. (d); The two given vertices are rational. In an equilateral trangle if two vertices are rational then the third vertex should be irrational only (d) option have irrational vertex.

24. (b); $(k-2)x + 3y = 4$

$(k-3)x - 2y = -5$

For being their lines are parallel.

$\frac{k-2}{k-3} = \frac{3}{-2}$

$-2k + 4 = 3k - 9$

$5k = 13, k = \frac{13}{5}$

25. (a); $2x + 3y - 6 = 0$

$x + ky + 1 = 0$

For perpendicular $a_1a_2 + b_1b_2 = 0$

$2 \times 1 + 3 \times k = 0$

$3k = -2$

$k = -\frac{2}{3}$

26. (c); $x + y + 3 = 0$ ... (i)

$x - y - 1 = 0$ ... (ii)

On solving (i) and (ii)

$x = -1, y = -2$

Line $ax + y + 4 = 0$ passes through $(-1, -2)$

$-a - 2 + 4 = 0, \quad a = 2$

27. (b); Distance of line form origin = $\frac{|c|}{\sqrt{a^2 + b^2}}$

Distance = $\frac{|c|}{\sqrt{4^2 + 3^2}} = \frac{9}{5}$

28. (c); Slope of the line is $\frac{3}{\sqrt{3}} = \sqrt{3}$

The inclination of the line 60°.

The required line is parallel to x axis.

Required equation of line $y = 1$

$y - 1 = 0$

29. (a); Point of intersection of lines $2x - 3y + 9 = 0$ and $x - y + 1 = 0$.

Point of intersection of the lines = (6, 7)

Required distance = $\sqrt{(6-2)^2 + (7-3)^2}$

$= \sqrt{4^2 + 4^2}$

$= 4\sqrt{2}$